

ENGR 102 Summer Bridge**Day 5 Instructor Key**

Compound Boolean Evidence and Complete Decisions

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as and, or, and not. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Compare two Boolean groupings.**5 points**

```

trained = False
temperature = 22
override = True

ready = trained and 15 <= temperature <= 30 or override
safe = trained and (15 <= temperature <= 30 or override)
print(ready, safe)

```

Evaluate each component and explain why the two stored results differ.

Answer and explanation

Python evaluates and before or. ready becomes (False and True) or True, which is True. safe requires trained on the outside, so False and (True or True) is False. Exact output: True False.

Question 2. Trace negation and a dependent readiness rule.**5 points**

```

flow_ok = True
label = "pending"
needs_review = not flow_ok or label != "approved"
ready = flow_ok and not needs_review
print(needs_review, ready)

```

Evaluate each Boolean in dependency order and give exact output.

Answer and explanation

not flow_ok is False, but label != approved is True, so needs_review is True. ready then combines True with not True, which is False. Exact output: True False.

Question 3. Program construction**10 points**

Write one complete Python program that satisfies every requirement.

- Read flow as a float, approval as text, and sensor as text.
- Print invalid if flow is negative.
- Otherwise print ready only when flow is from 18 through 22 inclusive, approval is exactly yes, and sensor is exactly ready.
- Print hold for every other nonnegative case. Print exactly one word.

Answer and explanation

```
flow = float(input())
approval = input()
sensor = input()

if flow < 0:
    print("invalid")
elif 18 <= flow <= 22 and approval == "yes" and sensor == "ready":
    print("ready")
else:
    print("hold")
```

The invalid-input branch comes first. The ready branch names all three required pieces of evidence with and, so no single passing condition can approve the record.

Protective tests include 18/yes/ready and 22/yes/ready as ready; 22.1/yes/ready and 20/no/ready as hold; and -1/yes/ready as invalid.

Scoring guidance

2 input/conversion, 2 invalid guard, 3 complete compound condition, 2 exclusive exact output, 1 boundary tests.